ADSP 31014 Statistical Models for Data Science
Course Project Part 2 Example

Business Problem

The Chicago Department of Transportation is interested in understanding how Chicago taxi trip duration is related to other factors of the trip.

Linear Regression Model

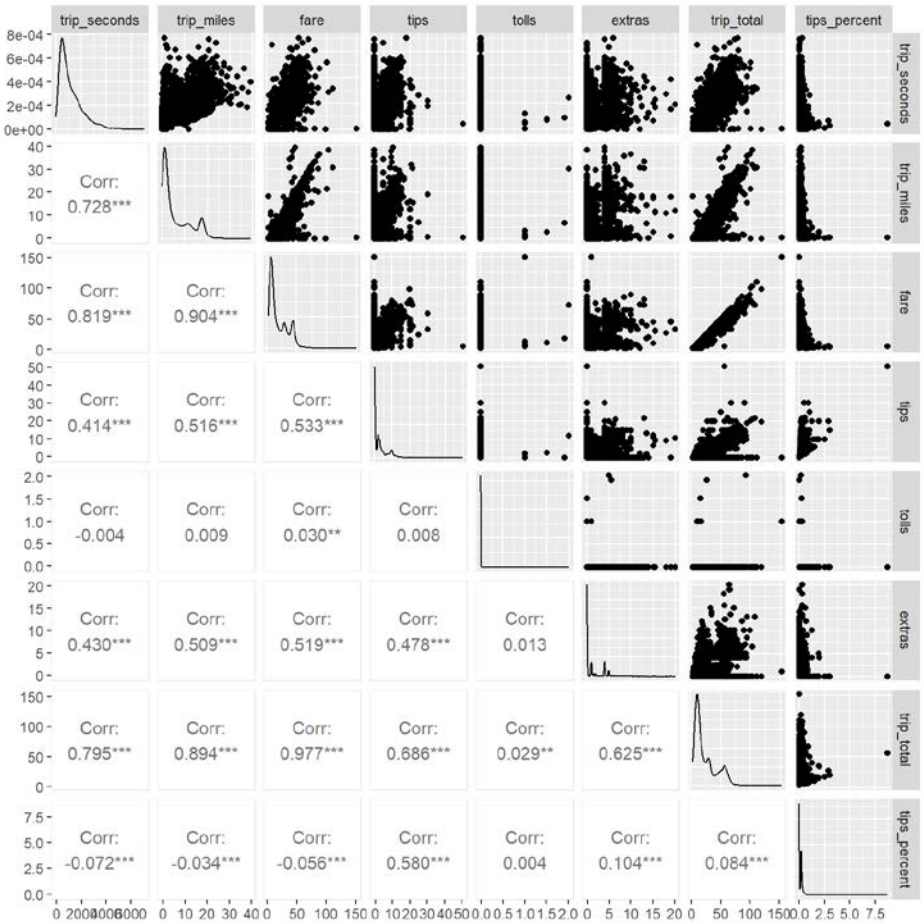
We build a linear regression model using a cleaned random sample of 100,000 historical Chicago taxi trip data from year 2022. We use trip_seconds as the response variable and other useful information as explanatory variables. The model formula is

```
log(trip_seconds) ~ log(fare) * factor(hour) + factor(month) + factor(day_of_week)
                  + pickup_area * dropoff_area + tips + tips_percent + extras
```

This model has 83 parameters with $R^2 = 0.74$. See Appendix 1 for model summary and estimated coefficients.

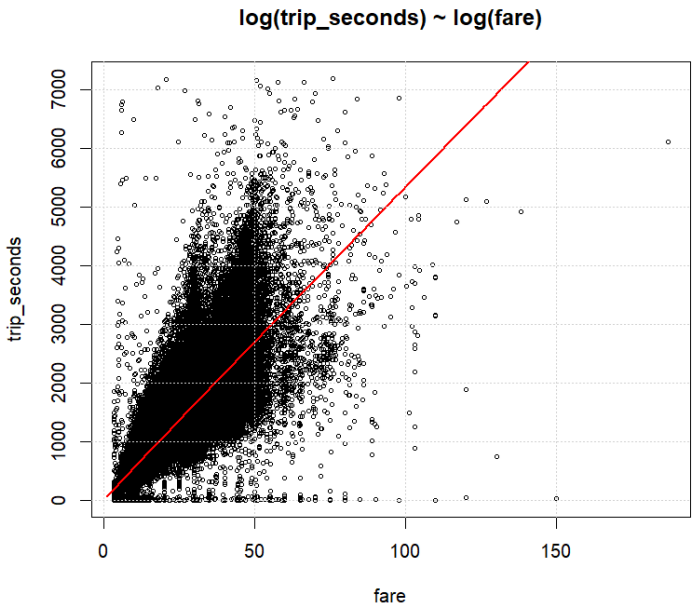
Key Considerations in Modeling

- All of trip_seconds, trip_miles, fare, and trip_total are highly correlated numerical variables.
- We focus on fare as the main explanatory variable with the highest correlation and omit trip_miles and trip_total to avoid multicollinearity and to improve explainability of the final model.

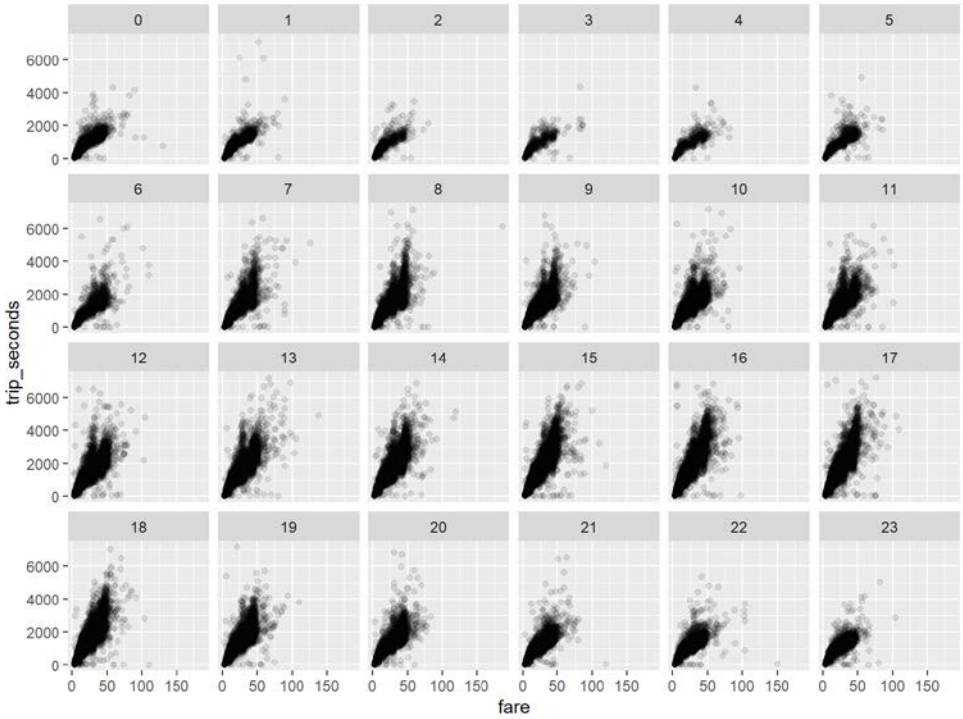


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- Variance of trip_seconds increases with fare i.e. heteroscedasticity. But otherwise, their relationship should be proportional. So we take logarithm of both. This baseline model has $R^2 = 0.70$.



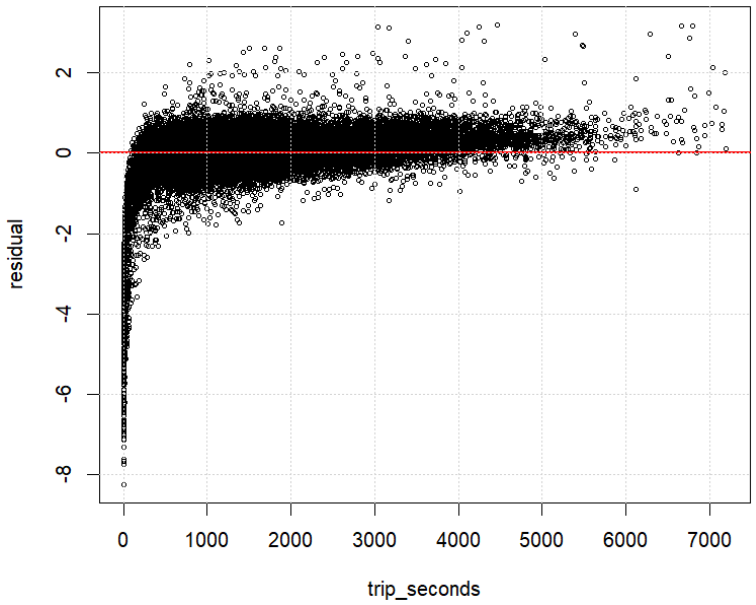
- Categorical hour interacts with fare. Month and day_of_week also seem useful.



- The model includes interaction between pickup_area and dropoff_area so that all $4 \times 4 = 16$ combinations are represented in the model.
- Among other numerical explanatory variables that are not highly correlated with trip_seconds, tolls is not useful (p-value 0.058) while trips, tip_percent, and extras are. This final model has $R^2 = 0.74$.

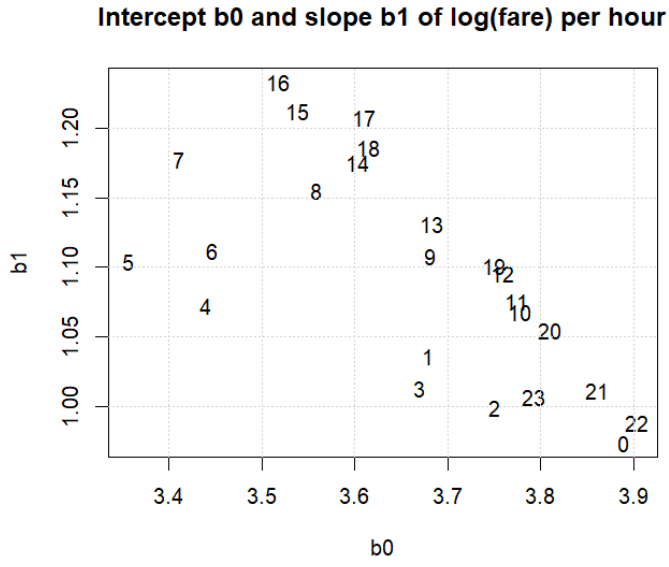
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- Additional interaction terms among the explanatory variables might improve R^2 slightly but would make the model overly-complicated.
- Although payment_type and company are significant in terms of F-test (p-value 0), it is likely an artifact of having a large data set. There is no reason to believe they are related to trip_seconds. They do not improve R^2 much.
- Plot of trip_seconds response vs residual shows the model is now closer to homoscedasticity. The over-estimation for short trip duration is due to outliers. It cannot be resolved by the linear regression model.



Other Insights

- The under-estimation for long trip duration is likely due to exogenous factors (such as traffic) that are not represented in our data set.
- To visualize the interaction between fare and hour, we plot the intercept and slope parameters per hour. There is a natural clustering of consecutive hours of the day. Rush hours tend to have larger slope parameters than midnight hours.



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Appendix 1

```
Call:
lm(formula = log(trip_seconds) ~ log(fare) + factor(hour) + factor(month) +
    factor(day_of_week) + pickup_area + dropoff_area + tips +
    tips_percent + extras + log(fare):factor(hour) + pickup_area:dropoff_area,
    data = data)

Residuals:
    Min      1Q  Median      3Q     Max
-8.2534 -0.1560  0.0491  0.2270  3.1926

Coefficients:
                                  Estimate Std. Error t value Pr(>|t|)
(Intercept)                      3.8900720  0.0449255  86.589  < 2e-16 ***
log(fare)                        0.9735510  0.0152825  63.704  < 2e-16 ***
factor(hour)1                   -0.2107594  0.0665295  -3.168 0.001536 **
factor(hour)2                   -0.1387306  0.0788807  -1.759 0.078625 .
factor(hour)3                   -0.2198479  0.0872176  -2.521 0.011714 *
factor(hour)4                   -0.4502049  0.0877282  -5.132 2.87e-07 ***
factor(hour)5                   -0.5334574  0.0818494  -6.518 7.18e-11 ***
factor(hour)6                   -0.4434281  0.0622606  -7.122 1.07e-12 ***
factor(hour)7                   -0.4793875  0.0534361  -8.971  < 2e-16 ***
factor(hour)8                   -0.3311573  0.0502697  -6.588 4.49e-11 ***
factor(hour)9                   -0.2089049  0.0497400  -4.200 2.67e-05 ***
factor(hour)10                  -0.1122178  0.0492824  -2.277 0.022786 *
factor(hour)11                  -0.1165457  0.0491923  -2.369 0.017829 *
factor(hour)12                  -0.1309277  0.0487473  -2.686 0.007236 **
factor(hour)13                  -0.2072340  0.0484988  -4.273 1.93e-05 ***
factor(hour)14                  -0.2865540  0.0484453  -5.915 3.33e-09 ***
factor(hour)15                  -0.3518976  0.0483736  -7.275 3.50e-13 ***
factor(hour)16                  -0.3729306  0.0482437  -7.730 1.08e-14 ***
factor(hour)17                  -0.2806625  0.0484287  -5.795 6.84e-09 ***
factor(hour)18                  -0.2758904  0.0486237  -5.674 1.40e-08 ***
factor(hour)19                  -0.1405302  0.0490943  -2.862 0.004205 **
factor(hour)20                  -0.0799337  0.0502611  -1.590 0.111755
factor(hour)21                  -0.0291401  0.0519490  -0.561 0.574842
factor(hour)22                   0.0135970  0.0541202   0.251 0.801632
factor(hour)23                  -0.0975745  0.0574447  -1.699 0.089401 .
factor(month)2                   0.0450446  0.0091197   4.939 7.85e-07 ***
factor(month)3                   0.0533534  0.0083654   6.378 1.80e-10 ***
factor(month)4                   0.0692966  0.0083613   8.288  < 2e-16 ***
factor(month)5                   0.0842447  0.0082105  10.261  < 2e-16 ***
factor(month)6                   0.0862150  0.0081171  10.621  < 2e-16 ***
factor(month)7                   0.0718850  0.0082818   8.680  < 2e-16 ***
factor(month)8                   0.0827699  0.0082143  10.076  < 2e-16 ***
factor(month)9                   0.0812746  0.0081653   9.954  < 2e-16 ***
factor(month)10                  0.0945059  0.0081316  11.622  < 2e-16 ***
factor(month)11                  0.0568868  0.0082960   6.857 7.07e-12 ***
factor(month)12                  0.0432809  0.0083113   5.207 1.92e-07 ***
factor(day_of_week)1             0.0313217  0.0054407   5.757 8.59e-09 ***
factor(day_of_week)2             0.0440304  0.0053909   8.168 3.18e-16 ***
factor(day_of_week)3             0.0589087  0.0053623  10.986  < 2e-16 ***
factor(day_of_week)4             0.0590660  0.0053554  11.029  < 2e-16 ***
factor(day_of_week)5             0.0274843  0.0056869   4.833 1.35e-06 ***
factor(day_of_week)6            -0.0546055  0.0059587  -9.164  < 2e-16 ***
pickup_areaMidway               -0.3236815  0.0148586 -21.784  < 2e-16 ***
pickup_areaO'Hare               -0.3323141  0.0107123 -31.022  < 2e-16 ***
pickup_areaOther                 0.0104056  0.0058687   1.773 0.076219 .
dropoff_areaMidway              -0.2375102  0.0164539 -14.435  < 2e-16 ***
```

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dropoff_areaO'Hare                    -0.3319426  0.0108509 -30.591  < 2e-16 ***
dropoff_areaOther                     -0.0021098  0.0049347  -0.428 0.668984
tips                                  -0.0143337  0.0007659 -18.716  < 2e-16 ***
tips_percent                           0.1860688  0.0120566  15.433  < 2e-16 ***
extras                                 0.0236595  0.0013697  17.274  < 2e-16 ***
log(fare):factor(hour)1                0.0624218  0.0231103   2.701 0.006913 **
log(fare):factor(hour)2                0.0254029  0.0280656   0.905 0.365402
log(fare):factor(hour)3                0.0389852  0.0299532   1.302 0.193077
log(fare):factor(hour)4                0.0987410  0.0287154   3.439 0.000585 ***
log(fare):factor(hour)5                0.1308029  0.0260079   5.029 4.93e-07 ***
log(fare):factor(hour)6                0.1386996  0.0207096   6.697 2.13e-11 ***
log(fare):factor(hour)7                0.2040174  0.0183964  11.090  < 2e-16 ***
log(fare):factor(hour)8                0.1820390  0.0175209  10.390  < 2e-16 ***
log(fare):factor(hour)9                0.1345343  0.0172406   7.803 6.09e-15 ***
log(fare):factor(hour)10               0.0942843  0.0170300   5.536 3.10e-08 ***
log(fare):factor(hour)11               0.1021021  0.0170293   5.996 2.03e-09 ***
log(fare):factor(hour)12               0.1219649  0.0168073   7.257 4.00e-13 ***
log(fare):factor(hour)13               0.1577934  0.0166991   9.449  < 2e-16 ***
log(fare):factor(hour)14               0.2022734  0.0166841  12.124  < 2e-16 ***
log(fare):factor(hour)15               0.2388200  0.0166665  14.329  < 2e-16 ***
log(fare):factor(hour)16               0.2597615  0.0166807  15.573  < 2e-16 ***
log(fare):factor(hour)17               0.2348008  0.0167675  14.003  < 2e-16 ***
log(fare):factor(hour)18               0.2124715  0.0168798  12.587  < 2e-16 ***
log(fare):factor(hour)19               0.1278705  0.0169974   7.523 5.40e-14 ***
log(fare):factor(hour)20               0.0812465  0.0173065   4.695 2.68e-06 ***
log(fare):factor(hour)21               0.0374562  0.0178807   2.095 0.036192 *
log(fare):factor(hour)22               0.0142928  0.0187356   0.763 0.445544
log(fare):factor(hour)23               0.0327988  0.0198289   1.654 0.098112 .
pickup_areaMidway:dropoff_areaMidway   0.0644057  0.0441931   1.457 0.145017
pickup_areaO'Hare:dropoff_areaMidway  -0.1463776  0.0565894  -2.587 0.009692 **
pickup_areaOther:dropoff_areaMidway    0.0210864  0.0273209   0.772 0.440232
pickup_areaMidway:dropoff_areaO'Hare  -0.1105393  0.0659974  -1.675 0.093956 .
pickup_areaO'Hare:dropoff_areaO'Hare   0.0387848  0.0223985   1.732 0.083352 .
pickup_areaOther:dropoff_areaO'Hare    0.0523918  0.0155596   3.367 0.000760 ***
pickup_areaMidway:dropoff_areaOther    0.0187916  0.0179465   1.047 0.295061
pickup_areaO'Hare:dropoff_areaOther    0.0498773  0.0094911   5.255 1.48e-07 ***
pickup_areaOther:dropoff_areaOther    -0.1469626  0.0073895 -19.888  < 2e-16 ***
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 0.4576 on 99917 degrees of freedom
Multiple R-squared:  0.7379,         Adjusted R-squared:  0.7377
F-statistic:  3430 on 82 and 99917 DF,  p-value: < 2.2e-16
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